llmXive Automated Review of GrepSeek: Training Search Agents for Direct Corpus Interaction

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is **advisory, automated feedback for the authors**: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The paper makes several specific factual claims regarding dataset composition, performance metrics, and statistical significance that require tighter alignment with their cited sources or internal consistency.

First, the description of the corpus in Section 3 ("Experimental Setup") and the Introduction states: "Corpus: 2018 Wikipedia (21M docs, 14GB)." This claim is attributed to \citep{karpukhin2020dense}. However, the Karpukhin et al. (2020) paper (DPR) describes a corpus of 21M *passages* extracted from Wikipedia, not 21M *documents*. Furthermore, the 14GB figure likely refers to the raw text size of the dump, whereas the DPR corpus is often processed and tokenized. Citing the DPR paper for the specific claim of "21M docs" and "14GB" is slightly inaccurate; the authors should clarify if they are using the raw 2018 dump or the DPR passage set and cite the appropriate source for the file size and document count statistics.

Second, there is a potential ambiguity in the latency claims. The Introduction states the sharded engine reduces latency from 5.39s to 0.71s, "bringing end-to-end latency to 8.6 seconds per query." The 0.71s figure clearly refers to the retrieval component. The jump to 8.6s implies the LLM generation and tool orchestration take approximately 7.9s. While this is plausible, the phrasing could be misinterpreted as the retrieval engine taking 8.6s. The text should explicitly clarify that the 8.6s is the *total* end-to-end time (retrieval + generation) to ensure the claim accurately reflects the contribution of the proposed engine versus the LLM.

Finally, the paper frequently marks results as statistically significant ($^\uparrow$) in Tables 1 and 2. The Appendix (Table 3 caption) mentions "McNemar's test, p<0.05" for the EM results.

However, the main text and Table 1 (F1 results) do not explicitly specify the statistical test used for the F1 significance claims. To support the ”significant” claims rigorously, the authors should confirm that the same statistical test (or an appropriate one for F1 scores) was applied to the F1 results and report this consistently in the main text or table captions.

- [**writing**] The paper makes several specific factual claims regarding dataset composition, performance metrics, and statistical significance that require tighter alignment with their cited sources or internal consistency. First, the description of the corpus in Section 3 (”Experimental Setup”) and the Introduction states: ”Corpus: 2018 Wikipedia (21M docs, 14GB).” This claim is attributed to \citep{karpukhin2020dense}. However, the Karpukhin et al. (2020) paper (DPR) describes a corpus of 21M *passages* ext

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1
Figure 1 effectively visualizes the architectural differences between Agentic RAG and Direct Corpus Interaction as described in the caption. The flowcharts are clear, the components are well-labeled, and the data flow is easy to follow without missing legends or illegible text.

Figure 2
Figure 2 effectively communicates the efficiency gains of the proposed method, but the caption contains a grammatical error. Additionally, the visualization of the ’Ours’ data in subplots (b) and (c) is suboptimal due to axis scaling that renders the significant improvements visually negligible.

Figure 3
Figure 3 effectively displays training dynamics across three metrics with clear axes, units, and a comprehensive legend. The visual data aligns perfectly with the provided caption, and the inclusion of error bands (shaded regions) adds necessary context to the mean trends.

Figure 4
The figure effectively visualizes the performance impact of SFT trajectory size, but the x-axis label is grammatically incomplete and the ’base (0)’ tick lacks a clear definition in the caption.

- [**writing**] Figure 2 caption: The phrase ’Efficiency and cost analysis of compared to’ contains a grammatical error (missing subject) and should be corrected to ’Efficiency and cost analysis of GrepSeek compared to’.
- [**science**] Figure 2b: The ’Ours’ bar represents 14 GB of RAM, which is significantly lower than the baselines (70 GB and 221 GB). The y-axis scale (0-200+) makes this bar appear negligible; a broken axis or inset would better visualize the magnitude of this improvement.
- [**science**] Figure 2c: The ’Ours’ bars are labeled ’1m’ (1 minute) but are plotted at the zero baseline, making them visually indistinguishable from zero. This obscures the data; the y-axis should be adjusted or the bars highlighted to show they are non-zero.
- [**writing**] Figure 4: The x-axis label ’# SFT Trajectory’ is ambiguous regarding the ’base

(0)' tick; the caption does not clarify if 'base' represents a model with zero SFT trajectories or a pre-trained baseline.

- **[writing]** Figure 4: The x-axis label '# SFT Trajectory' is grammatically incomplete and should be pluralized to '# SFT Trajectories' to match the caption.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript relies heavily on domain-specific acronyms and technical shorthand that are not defined upon first use, creating a barrier for non-specialist readers. The term "DCI" (Direct Corpus Interaction) is central to the paper's contribution but is never explicitly expanded in the Abstract or Introduction, appearing only as an acronym. Similarly, "GRPO" (Group Relative Policy Optimization) and "SFT" (Supervised Fine-Tuning) are used frequently in Sections 2 and 3 without initial definition. The acronym "OOD" (out-of-distribution) appears in Section 3.1 without expansion. Additionally, "FSDP" is mentioned in the context of the training framework without explanation. While these terms are standard in the immediate sub-field of LLM training and retrieval, the paper's goal of demonstrating a novel interaction paradigm suggests a need for broader accessibility. The authors should ensure every acronym is spelled out at its first occurrence in the main text. Furthermore, phrases like "sharded-parallel execution engine" and "semantics-preserving" are dense; a brief, plain-language elaboration of the mechanism (e.g., "splitting the data into parts to run simultaneously") would significantly improve readability for a general computer science audience.

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paper_reviewer_logical_consistency

Verdict: minor_revision
The logical flow regarding the efficiency gains in Section 2.2 ("Efficient Corpus Interaction") contains a potential conflation of variables. The text states that sharded-parallel execution reduces latency from 5.39s to 0.71s. However, the description of the "Persistent Search Daemon" (which keeps the corpus in memory) is presented as a separate bullet point. It is logically ambiguous whether the 5.39s baseline includes the overhead of process startup for every query (which the daemon eliminates) or if it represents a pure sequential execution time. If the baseline includes startup overhead, the 7.6x speedup is not solely attributable to "sharded-parallel execution" as claimed, but rather a combination of parallelism and daemonization. The causal claim that "sharded-parallel execution... accelerating retrieval by up to 7.6x" is therefore not

fully supported by the stated premises without isolating the daemon's contribution.

Furthermore, the argument that the method ”excels at exact entity matching... where dense retrievers fail” relies heavily on the qualitative case studies (e.g., the chemical formula example). While these examples are valid, the leap to the general conclusion that this mechanism drives the *overall* multi-hop performance gain (Avg F1 0.5691) is not rigorously supported. The paper does not provide a breakdown of *which* specific questions in the multi-hop datasets were solved via ”exact entity matching” versus other reasoning paths. Without this, the causal link between the specific mechanism (lexical precision) and the aggregate metric improvement remains an assumption rather than a demonstrated fact.

Finally, the ablation study (Table 2) shows that removing SFT leads to a catastrophic performance drop (Avg F1 0.5691 -> 0.3314). The text attributes this to ”unstable behavior” in direct RL. However, the logic of the ”Cold-Start” data generation (Section 2.1.1) posits that the Tutor/Planner generate ”verified, causally grounded trajectories.” If these trajectories are truly verified and grounded, the logical expectation is that the model should be able to learn the policy from them even with a smaller SFT set or potentially via RL alone if the reward signal is strong. The massive dependency on SFT suggests that the ”verified” trajectories might not be sufficient to bootstrap the policy without the specific distributional matching of SFT, a nuance that the current argument glosses over. The claim that ”Direct RL on large corpora causes unstable behavior” is a premise, but the evidence (the ablation) only shows that *without SFT*, performance is low; it does not explicitly prove that the instability is the *cause* of the low performance rather than a lack of convergence or reward sparsity.

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paper_reviewer_overreach

Verdict: minor_revision

The paper makes several strong claims regarding the superiority and practicality of Direct Corpus Interaction (DCI) that slightly overreach the presented evidence, particularly regarding the trade-offs between latency, surface-form robustness, and the definition of ”indexing.”

First, the narrative in Section 3.2 (”Main Findings”) claims the method ”outperforms all baselines on 4/7 datasets” and frames the performance on PopQA as a ”minor degradation.” However, Table 1 explicitly marks the PopQA result (0.4861) with a down-arrow (\$\downarrow\$), indicating a statistically significant drop compared to Search-R1 (0.5101). Given that the paper's core argument relies on the agent's ability to handle complex retrieval, a significant failure on a standard benchmark due to diacritic sensitivity (as admitted in the text) suggests the method is not a universal replacement for dense retrieval. The conclusion should more honestly frame this as a trade-off: superior precision on exact matches at the cost of robustness to surface variations, rather than a general ”outperformance.”

Second, the efficiency claim of ”zero offline indexing time” (Section 3.2) is technically accurate regarding the *retrieval index* but misleading in the broader context of the system’s cost. The ”Cold-Start Data Generation” phase (Section 2.1) involves a computationally expensive process: using a 27B Tutor model and a Planner to generate 10,000 verified trajectories. This pre-computation is a form of heavy lifting that replaces the traditional index build. While the authors correctly note they avoid building a vector database, characterizing the entire pipeline as having ”zero offline” costs ignores the substantial compute required to synthesize the training data. The claim should be refined to ”zero offline *retrieval* indexing” to avoid implying the method is computationally free to deploy.

Finally, the conclusion asserts the method is a ”practical alternative” (Section 5). While the memory savings (14GB vs 221GB) are impressive, the latency is significantly higher (8.67s vs 4.77s for E5). For many real-time search applications, a near-doubling of latency is a critical barrier. The paper overreaches by presenting the method as broadly practical without explicitly qualifying that its practicality is contingent on specific constraints (e.g., memory-limited environments or scenarios where exact string matching is paramount over speed). The discussion should more clearly delineate the specific use-cases where this latency penalty is acceptable.

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paper_reviewer_safety_ethics

Verdict: minor_revision

The paper presents a Direct Corpus Interaction (DCI) agent trained via a two-stage process involving a ”Tutor” model that is explicitly ”answer-aware” (Section 2.1.1, Appendix). From a safety and ethics perspective, the primary concern is the potential for data leakage and the integrity of the evaluation. The authors state that the Tutor decomposes queries and answers to construct retrieval chains in reverse. It is critical to explicitly confirm in the text that the evaluation datasets (NQ, HotpotQA, etc.) were strictly held out from the Tutor’s training data and that the ”answer-leak rule” effectively prevents the synthetic training trajectories from containing memorized ground-truth answers that could artificially inflate performance metrics. Without this explicit confirmation, there is a risk that the reported gains are due to overfitting on test data rather than genuine reasoning capabilities.

Additionally, the methodology relies on the agent executing shell commands (e.g., ‘grep‘, ‘rg‘) directly against a raw corpus. While the system prompt (Figure 1) and the sharded-parallel engine (Algorithm 1) impose constraints like ”No redirection,” the paper should briefly address the security of the execution environment. Specifically, how is the system sandboxed to prevent the agent from being manipulated (via prompt injection) to execute arbitrary commands or access files outside the intended corpus? Given the dual-use potential of training agents to interact directly with file systems, a brief statement on the containment measures (e.g., read-only mounts, restricted user privileges) would strengthen the ethical rigor of the work.

Finally, the ”answer-leak rule” is described as masking target entities, but the robustness of this masking against the Tutor model’s ability to infer or hallucinate the answer from context is not detailed. A sentence clarifying the verification process for the synthetic trajectories would alleviate concerns about the quality and safety of the training data generation pipeline.

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paper_reviewer_scientific_evidence

Verdict: minor_revision

The paper presents a compelling empirical case for Direct Corpus Interaction (DCI) using shell commands, supported by extensive benchmarking across seven datasets. The central claim—that a compact agent can outperform dense retrievers on multi-hop tasks by avoiding semantic conflation—is well-supported by the reported F1 and Exact Match scores (Table 1, Table 2). The ablation studies (Table 1, Table 2) effectively demonstrate the necessity of both the SFT cold-start and the GRPO refinement stages, with the full model showing consistent gains over variants.

However, the scientific evidence requires minor strengthening regarding statistical rigor and experimental transparency. First, the manuscript repeatedly asserts ”statistically significant improvement” (denoted by $^\uparrow$) with $p<0.05$ but does not specify the statistical test employed (e.g., McNemar’s test, paired t-test, or bootstrap confidence intervals) nor the number of independent runs (seeds) used to generate the mean scores. Given the stochastic nature of RL training and the large evaluation sets, reporting the variance or the specific test used is essential to validate these significance claims.

Second, the efficiency comparison, while a key contribution, conflates runtime memory usage with index storage size. The claim that the method uses ”14GB RAM” versus ”221GB for dense indices” (Section 3.2) compares the agent’s active memory footprint against the pre-computed index size of baselines. To fully substantiate the ”zero offline indexing” advantage, the authors should clarify if the 8.67s latency includes the initial corpus loading time (warm-up) and whether the 14GB figure represents peak memory during the search or the static corpus size.

Finally, the ablation study for the ”w/o GRPO” condition (Table 1) shows the model still outperforms most baselines. The text implies this is a ”drastic reduction,” but the magnitude of improvement over the SFT-only baseline varies by dataset. A brief discussion on whether the GRPO stage primarily improves robustness or just final accuracy on specific hard cases would strengthen the interpretation of the RL contribution. Overall, the evidence is robust, but these clarifications are necessary to fully validate the statistical and efficiency claims.

- [**science**] The claim of statistical significance (p<0.05) for F1 and EM improvements lacks methodological detail. Specify the statistical test used (e.g., McNemar’s, bootstrap) and report the number of random seeds or runs used to estimate variance, as single-run results on large datasets can be noisy.
- [**science**] The ablation study (Tables 1 & 2) shows a massive performance drop when removing SFT, but the ’w/o GRPO’ variant still outperforms baselines on some metrics. Clarify if the GRPO-only baseline was trained from scratch or initialized from the SFT model, as this affects the interpretation of the ’cold-start’ contribution.
- [**science**] The efficiency claim (14GB RAM vs 221GB) compares the agent’s runtime memory against the index size of dense retrievers. To support the ’zero offline indexing’ claim robustly, explicitly state the time required to load the 14GB corpus into RAM and whether this ’warm-up’ cost is included in the reported 8.67s latency.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The statistical analysis presented in the manuscript is currently insufficient to support the strong claims of ”statistically significant improvement” made throughout the results section.

First, while Tables 1 and 2 (F1 and EM scores) mark specific improvements with a superscript $\uparrow$ and the caption notes $p<0.05$, the text fails to specify which statistical test was employed (e.g., McNemar’s test for paired classification, paired t-test, or bootstrap). Furthermore, the number of independent experimental runs (random seeds) used to generate these point estimates is not reported. In reinforcement learning experiments, performance can vary significantly based on initialization; reporting a single run’s result as ”significant” without variance estimates or a defined null hypothesis test is methodologically unsound.

Second, the experimental setup describes a training process of only 200 GRPO steps. While the ablation study (Tables 1 and 2) demonstrates that removing SFT or GRPO degrades performance, it does not provide confidence intervals or standard deviations for these comparisons. The magnitude of the drop (e.g., Avg F1 from 0.5691 to 0.4249) appears large, but without a formal test or error bars, the robustness of this conclusion remains unverified.

Finally, the efficiency claims (latency reduction from 5.39s to 0.71s) are presented as deterministic facts. While the sharded engine logic is deterministic, the end-to-end latency in a distributed system often has variance. If these are single measurements, they should be presented as such, or accompanied by standard deviation if multiple runs were performed.

To resolve these issues, the authors must: (1) explicitly state the statistical test used for significance claims and the number of seeds/runs averaged; (2) report standard deviations or 95% confidence intervals for all primary metrics; and (3) apply the same statistical rigor to the ablation comparisons.

- [**science**] The paper claims statistical significance ($p<0.05$) for performance gains in Tables 1 and 2 (F1 and EM) but does not specify the statistical test used (e.g., McNemar's, paired t-test) or the number of independent runs. Without variance estimates or test details, these significance claims are unverifiable.
- [**science**] The RL training protocol specifies only 200 steps with a group size of $n=5$. The paper reports single-point performance metrics without confidence intervals or standard deviations across multiple random seeds, making it impossible to assess the stability or reproducibility of the reported gains.
- [**science**] The ablation study (Tables 1 & 2) compares the full model against variants without reporting statistical significance for the differences between the ablated models and the baseline. It is unclear if the observed drops are statistically significant or within the noise of the evaluation.

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript demonstrates a generally high standard of academic writing, with clear structure and logical flow throughout the main sections. The introduction effectively sets the stage for the Direct Corpus Interaction (DCI) approach, and the experimental results are presented with appropriate detail. However, there are a few areas where clarity could be improved to ensure the text is accessible to a broader audience within the field.

In Section 2.1, the terms "answer-aware Tutor" and "answer-blind Planner" are used to describe the components of the cold-start data generation pipeline. While the context implies their functions, the specific meaning of "answer-aware" and "answer-blind" is not explicitly defined. A brief clarification, such as "a Tutor that has access to the ground-truth answer" and "a Planner that does not," would eliminate potential ambiguity for readers unfamiliar with the specific terminology.

In Section 3.2, the discussion of the model's strengths mentions that it "excels at exact entity matching and rare patterns... where dense retrievers fail due to semantic conflation." The term "semantic conflation" is used here to explain the failure mode of dense retrievers, but its precise meaning in this context is not elaborated. Does it refer to the merging of distinct entities with similar embeddings, or the inability to distinguish between a specific entity and a broader category? A more precise description of this failure mode would strengthen the argument.

Additionally, in the Appendix (Section A.3), the reward function definitions rely on variables like $\mathcal Y$ and $\hat y$ without explicit definition in the immediate vicinity of the formula. While these are standard notations in the field, providing a brief definition (e.g., "where $\mathcal Y$ is the set of gold answers and $\hat y$ is the predicted answer") would enhance the self-containment and clarity of the appendix.

Overall, the writing is strong, but these minor clarifications would further improve the readability and precision of the manuscript.

- [**writing**] In Section 2.1 (Training DCI Search Agent), the phrase 'answer-aware Tutor' and 'answer-blind Planner' are introduced without defining what 'answer-aware' or 'answer-blind' specifically entails in this context. Clarify these terms or provide a brief parenthetical explanation to ensure reader comprehension.
- [**writing**] In Section 3.2 (Main Findings), the sentence 'It excels at exact entity matching and rare patterns (e.g., chemical formulas) where dense retrievers fail due to semantic conflation' is slightly ambiguous. Specify whether 'semantic conflation' refers to the retriever's inability to distinguish similar entities or its tendency to merge distinct concepts.
- [**writing**] In the Appendix (Section A.3, Reward Function), the formula for R_{ans} uses $\max_{y\in\mathcal Y}F_1(\hat y,y)$, but the text does not explicitly define $\mathcal Y$ (the set of gold answers) or $\hat y$ (the predicted answer) in this specific context, assuming prior knowledge. Define these variables for clarity.